PHYSICS | ADAPTLY

ELECTRICITY – CIRCUITS, CURRENT & RESISTANCE

KEY VOCABULARY

Current (I): the rate of flow of charge, measured in amperes (A).

Potential difference (V): energy transferred per unit charge, in volts (V).

Resistance (R): how much a component opposes the current, in ohms (Ω).

Charge (Q): measured in coulombs (C); $Q = I \times t$.

COMMON MISCONCEPTIONS

~~Current is used up by components.~~

✓ Current is the same all around a series circuit.

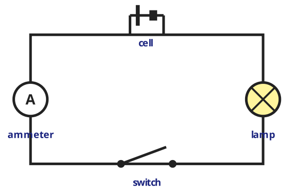
~~An ammeter is connected in parallel.~~

✓ An ammeter is connected in series.

~~More resistance means more current.~~

✓ More resistance means less current (for fixed V).

A SIMPLE SERIES CIRCUIT



Ammeter in **series**; voltmeter would go in **parallel**.

1 OHM'S LAW

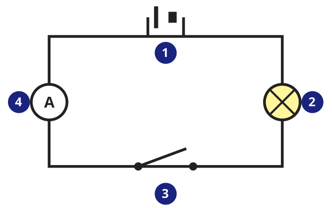
A resistor has a p.d. of 6 V and a current of 0.5 A. **Calculate** its resistance. **[2 marks]**

Answer: ____________________ Ω

2 NAME THE COMPONENTS

Name circuit components **1-4** from their symbols. **[4 marks]**

cell | lamp | switch | ammeter



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3 CHARGE FLOW

A current of 3 A flows for 120 s. **Calculate** the charge that flows ($Q = I \times t$). **[2 marks]**

Answer: ____________________ C

4 SERIES VS PARALLEL

Explain one advantage of connecting lamps in parallel rather than series. **[2 marks]**

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5 MEASURING

Complete the sentence using the word bank. **[2 marks]**

An _____ is placed in series; a voltmeter is placed in _____.

series | parallel | ammeter | voltmeter

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6 RESISTOR I–V GRAPH

Describe the I–V graph for a fixed resistor at constant temperature. **[2 marks]**

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TIP: *V = I × R. Rearrange to find any one when you know the other two.*

✓ **Think: Why does the resistance of a filament lamp increase as it gets hotter?**